Large Document Processing - A Judicial Document Case Study

Comprehensive Report
for pursuing

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by

A Aparajitha
2022814001
aparajitha.allamraju@research.iiit.ac.in

Advisor: Dr. Manish Shrivastava



Language Technologies Research Center
International Institute of Information Technology Hyderabad
500 032, India

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Abstract

This report aims to present a comprehensive analysis of the current research related to understanding and processing large connected documents. As a case study, the research is focused on Indian Judicial Documents. The legal domain brings in it's own set of challenges to supplement the existing challenges in Large Language Models (LLMs) and Text-Attributed Graphs (TAGs). The current work is focused on the use of LLMs for a judicial citation network. This report focuses on the many aspects of the problem by separately studying current research in LLMs, Graphs, Documents, and Legal NLP.

The current generation of LLMs, GPT-4, LLaMA2, PaLM to name a few, have shown good prowess in many complex tasks. Though their ability to solve many complex tasks has improved over older models they still have some limitations. These limitations are not only in the ability to perform tasks but also in infrastructural capabilities and needs. The report explores these limitations and some advances to solve these problems.

The report also explores the current work in the area of Graph Neural Networks. Since, the nature of the available data can be represented as a knowledge graph with the case documents/paragraphs as nodes; a connection between LLMs and Graphs is also explored.

As the work involves judicial documents downloaded from the various courts of India, the report includes some work in the domain of document understanding. Current work in the domain of Legal NLP is also included in the report.

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Chapter 1

Introduction

Pre-trained Large Language Models like BERT[1], PaLM[2], LLaMA[3], GPT[4] based on the Transformer architecture[5] have achieved state-of-the-art results in several downstream tasks and have become mainstream given their current popularity. As the size of the models has grown, certain inherent behaviour has also been observed. This has led to harnessing their vast knowledge through newer and existing paradigms such as zero-shot inference, few-shot learning, prompt tuning [6], reasoning [7], [8] and instruction tuning[9].

Despite the innate capabilities of LLMs, they also exhibit some limitations. One of the major limitation is dealing with longer sequences. The self-attention module of transformers has quadratic time and memory complexity making LLMs slow and memory hogging as sequence length increases. Sparse-approximation[10], [11], Low-rank approximation[12]–[14] and Flash Attention[15] have been proposed for FLOP and wall-clock reduction. SwitchTransformers [16] proposed a sparsely-activated expert model to attain greater computational efficiency.

The quadratic memory complexity is one of the reasons why LLMs cannot be used off the shelf on large text documents. Longformer[17] addresses this limitation through a linear scaling attention mechanism by using sliding window, dilated sliding window and global attention patterns. Real-world documents are not like the plain text documents used in NLP tasks. Documents 1) have a semantic structure, 2) may contain semi structured elements like forms and tables, 3) may contain images and charts and 4) have a spatial layout. LayoutLM[18] and UDoc [19] work on extending transformers to train across the various modalities.

Transformers are also limited in their understanding of graphs. Graphs are non-linearly structured and contain important information in their nodes and edges. Graphormer[20] introduces centrality encoding and spatial encoding to the transformer model to enable better representation of graphs.

As a case study here and research direction, I am working on document understanding with Indian Judicial documents. Similar to scientific documents, judicial documents can also be represented using a citation network. Utilizing LLMs for legal citation network is a relatively unexplored research area.

LegalBERT[21] explored BERT-based models - finetuned, continued pretraining and pretrained from scratch on US and EU Legal data to evaluate legal domain understanding. Lawformer[22] is a Longformer-based model that attempts to extend the context size further and is trained on Chinese Legal Documents.

InLegalBERT [23] is trained on ILDC [24]. HLDC[25] and MildSUM[26] are some other existing Indian Legal Datasets. LexGLUE[27] is a benchmark evaluation dataset on seven Legal tasks from the US and EU.

The report will expand on the mentioned limitations and current approaches to circumvent them. This report will also discuss current research in Document Understanding, Graphs and Legal NLP - models and evaluation.

1.1 Background of Indian Judiciary

Indian legal systems majorly follows common law with a few exceptions. This makes identifying and retrieving of case precedents important for lawyers and judges, but it is usually a time consuming process. Lawyers find similar cases to cite while preparing the arguments, whereas judges cite preceding cases while delivering the judgement. Precedents follow a hierarchy, based on the level of the court, year of judgement and size of the bench passing the judgement. Cases from Supreme Court have the highest priority followed by the State High Court, then the High Courts of other states and finally lower courts. If there is a precedent from a higher court, it cannot be usually overruled by a lower court.

A case may cite multiple precedents, each for similar aspects or different. Like-wise, a precedent might be cited by different cases for the same or different aspect. The context surrounding the citation can give insight into the citation aspects.

In the Indian Judicial System, while the Supreme Court of India and the State High Courts function in English, the regional courts may operate in regional languages. This multilingual nature added to the complex structure of legal documents leads to some unique and interesting problems that can be tackled using NLP.

Most judicial documents use unnecessarily complex sentences and inconsistent punctuation. There are also non-English words that are very specific to English Indian Legal Documents such as *Tehsildar*. The complex language structure carries over to the Legal Documents in regional languages. The regional documents contain a lot of jargon and word senses that are vastly different from their meaning in regular use.

1.2 Structure of the report

Section 2 talks in detail about some of the papers mentioned here. The first Section 2.1 elaborates on LLMs - their capabilities and limitations. Section 2.2 discusses two approaches for computational efficiency in LLMs. Section 2.3 discusses the research in the area of document understanding and multi-modal pretraining. Section 2.4 is about the current state of the art Graph Neural Networks (GNNs). It also attempts to include recent advances in using LLMs in conjugation with Graphs. Finally, Section 2.5 explores the current work in Legal LLMs and datasets.

Section 3 reports the current work progress in the research direction and Section 4 concludes with the findings in the work done so far and talks about the future direction of work.

Chapter 2

Literature Survey

2.1 LLMs - Tuning

The advent of hundreds-of-billions parameter LLMs, has given rise to their utility in zero-shot reasoning[7], Chain of Thought Reasoning[8] and Prompt Tuning[6]. These approaches allow us to harness the capabilities of LLMs without having to change the model parameters. This allows for model re-usability without having to store a separate finetuned model for each downstream task.

2.1.1 Zero-Shot Reasoning

[1] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," *Advances in neural information processing systems*, vol. 35, pp. 22 199–22 213, 2022.

LLMs have shown to be capable of in-context learning through few and zero shot examples. This is also known as *prompting*. In a few-shot setting, the model is conditioned with a few examples of the task. In zero-shot, the prompting is an instruction describing the prompt. LLMs have struggled with tasks that require multi-step reasoning like arithmetic.

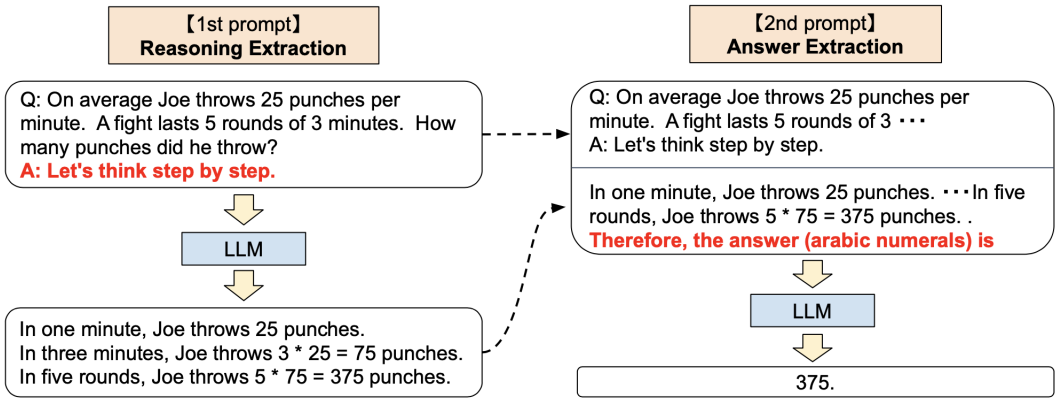


Figure 2.1 Robustness study against templates measured on the MultiArith dataset

No.	Category	Template	Accuracy
1	instructive	Let's think step by step.	**78.7**
2		First, (*1)	77.3
3		Let's think about this logically.	74.5
4		Let's solve this problem by splitting it into steps. (*2)	72.2
5		Let's be realistic and think step by step.	70.8
6		Let's think like a detective step by step.	70.3
7		Let's think	57.5
8		Before we dive into the answer,	55.7
9		The answer is after the proof.	45.7
10	misleading	Don't think. Just feel.	18.8
11		Let's think step by step but reach an incorrect answer.	18.7
12		Let's count the number of "a" in the question.	16.7
13		By using the fact that the earth is round,	9.3
14	irrelevant	By the way, I found a good restaurant nearby.	17.5
15		Abrakadabra!	15.5
16		It's a beautiful day.	13.1
-		(Zero-shot)	17.7

Figure 2.2 Robustness study against templates measured on the MultiArith dataset

This work shows how LLMs are decent zero-shot reasoners by simply adding *Let's think step by step* to the prompt fed to an instruction model - InstructGPT(text-davinci-002)[9]. The model was evaluated against several benchmark reasoning tasks including arithmetics (MultiArith[28], GSM8K[29], AQUA-RAT[30], SVAMP[31]), symbolic reasoning (Last Letter, Coin Flip), and other logical reasoning tasks (Date Understanding, Tracking Shuffled Objects). The major advantage of this approach is that this is inherently task-agnostic and can reason across a wide range of tasks with a single simple template.

The work uses prompting twice - first to get the model to reason and second to extract the answer from the reasoned output as shown in figure 2.1. To elicit reasoning, the first template used is "Q: [X]. A: [T]", where [X] is the slot for input question and [T] for a trigger sentence such as *Let's think step by step*. The list of triggers used in the first step can be seen in Figure 2.2. In the second step, "[X'] [Z] [A]", [X'] is the 1st prompt, [Z] is the reason generated in first step, and [A] for a trigger sentence to extract answer, is used as the template. The list of triggers used for the second prompte can be seen in Figure 2.3.

2.1.2 Chain of Thought Prompting

[1] J. Wei, X. Wang, D. Schuurmans, *et al.*, "Chain-of-thought prompting elicits reasoning in large language models," *Advances in Neural Information Processing Systems*, vol. 35, pp. 24 824–24 837, 2022.

While [7] showed the power of zero-shot reasoning through the use of a simple template, this work focuses on using few-shot prompting to elicit Chain-of-Thought (CoT) reasoning in LLMs. This makes

No	Task	Zero-Shot-CoT (left)	Zero-Shot-CoT (right)
1	SingleEq	Therefore, the answer (arabic numerals) is	Therefore, the answer is
2	AddSub	Therefore, the answer (arabic numerals) is	Therefore, the answer is
3	MultiArith	Therefore, the answer (arabic numerals) is	Therefore, the answer is
4	GSM8K	Therefore, the answer (arabic numerals) is	Therefore, the answer is
5	AQUA-RAT	Therefore, among A through E, the answer is	Therefore, the answer is
6	SVAMP	Therefore, the answer (arabic numerals) is	Therefore, the answer is
7	C.S.QA	Therefore, among A through E, the answer is	Therefore, the answer is
8	StrategyQA	Therefore, the answer (Yes or No) is	Therefore, the answer is
9	D.U.	Therefore, among A through F, the answer is	Therefore, the answer is
10	S.O.	Therefore, among A through C, the answer is	Therefore, the answer is
11	Last Letters	Therefore, the answer is	Therefore, the answer is
12	Coin Flip	Therefore, the answer (Yes or No) is	Therefore, the answer is

Figure 2.3 Answer extraction prompts used for Zero-shot-CoT experiments

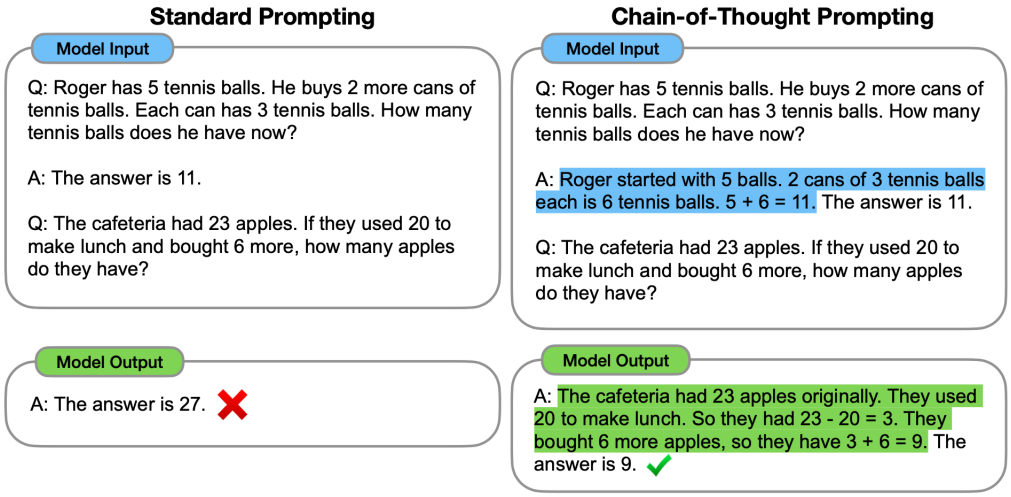


Figure 2.4 Example of reasoning with Chain of Thought prompting in an arithmetic task

the model generate intermediate reasoning steps in multi-hop reasoning problems. Figure 2.4 shows an example of the reasoning generated by CoT. The results show improvement is tasks like arithmetic, commonsense reasoning and symbolic reasoning.

There are four properties of CoT prompting that make it very attractive for use

1. CoT provides an interpretable window into the model's behaviour by showing it's reasoning path.

2. Additional computation can be allocated to problems that require more reasoning steps since the model allows for problem decomposition.

3. CoT can be used for any task that can be solved by a human via reasoning with language.

4. CoT is readily elicited in sufficiently large models, off the shelf through few-shot CoT prompting. No finetuning is required.

Chain of Thought Prompting was evaluated on five math word problem benchmarks, five commonsense reasoning benchmarks and two symbolic reasoning datasets using GPT3[32], LaMDA[33], PaLM[2], UL2 20B[34] and Codex[35]. Several ablations were carried out for the three tasks to draw up conclusions.

Summary of Results

- CoT prompting did not positively impact performance for small models, and showed significant gains when used with substantially large models of around 100B parameters like GPT-3 175B, LaMDA 137B and PaLM 540B.
- Chain of Thought prompting showed better performance gains for more complex arithmetic tasks.
- PaLM 540B and Codex showed the most performance gains in commonsense reasoning.
- PaLM 540B shows high results in the two toy symbolic reasoning tasks.

Limitations

- There is no guarantee of the correct reasoning path leading to the correct answer.
- CoT reasoning emerges only in large scale models making it costly to serve any real world purpose.

2.1.3 Prompt Tuning

[1] B. Lester, R. Al-Rfou, and N. Constant, "The power of scale for parameter-efficient prompt tuning," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021, pp. 3045–3059.

Like the previous two methods use, this paper also uses a frozen language model to perform downstream tasks. Unlike human annotated hard prompts, this paper introduces prompt tuning through soft prompts. Soft prompts can be learnt through backpropagation tuned with signals from labeled examples. The prompts guide the model to the task. The paper runs experiments with T5[36].

Prompting is done by prepending tokens to an input X to maximize the likelihood of the correct generation, Y with fixed model parameters θ i.e. $Pr_\theta(Y|[P;X])$. In Prompt Tuning, a fixed prompt of special tokens is used. Only the embedding of these special tokens can be updated. The conditional probability is now $Pr_{\theta,\theta_P}(Y|[P;X])$. The likelihood of Y is maximized via backpropagation while applying the gradients to θ_P. Prompt tuning enables the prompt to have its own set of parameters that can be updated. By adding a new input representation, prompt tuning modifies the model behaviour on how the input is processed.

Advantages of Prompt Tuning

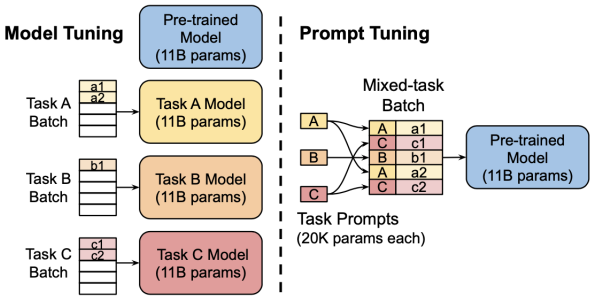


Figure 2.5 Prompt tuning only requires storing a small task-specific prompt for each task

- Since the model weights are frozen and only prompt is updated, task-specific prompts can be stored instead of task separate finetuned models as shown in Figure 2.5
- The performance of prompt tuned models has shown to rival the performance of finetuned models in the SuperGLUE[37] benchmark.
- Enables mixed-task inference using the original model.

Drawbacks of Prompt Tuning

- The prompt less is interpretable since it is in the embedding space unlike CoT prompting[8] that was previously explored.

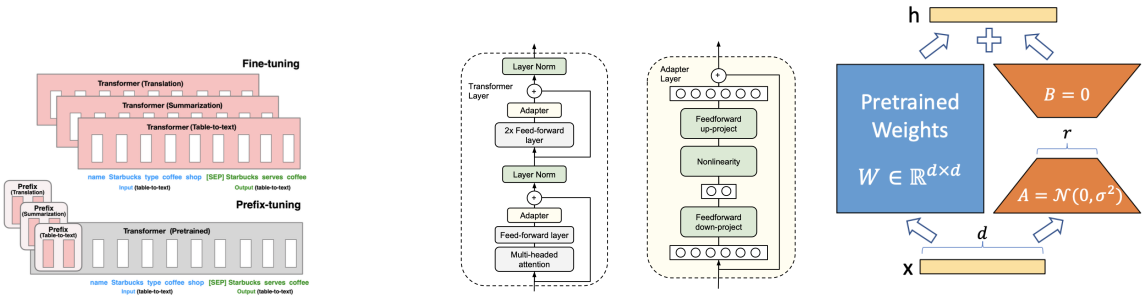
2.1.4 Other Related Work

The more conventional approaches like finetuning and continued pretraining on models of large sizes are not feasible for most people and many a times unnecessary. This has lead to several approaches using Parameter-Efficient Finetuning(PEFT) of large models, where only a few parameters of the model are trained. Prompt Tuning is one such technique which adds embeddings at the input level.

Prefix Tuning [38], another PEFT technique, uses a frozen model and adds a trainable prefix to every layer of the transformer block. Another PEFT method, involves adding Adapter layers in the transformer block to reduce the number of trainable parameters. Low Rank Adaptation (LoRA)[39] improves training efficiency by running calculations on lower rank matrices. It is built on the fact that over-parameterized models have a low intrinsic rank.

2.2 LLMs - Computational Efficiency

Flash Attention[15] and Switch Transformers[16] improve the computational efficiency of training a large model through making I/O aware decisions and using Mixture-of-Experts respectively.



(a) (b) (c)

Figure 2.6 PEFT Techniques (a) Finetuning vs Prefix Tuning (b) Adapter layers in Transformer Block (c) LoRA A, B are the trainable matrices

2.2.1 Flash Attention

[1] T. Dao, D. Fu, S. Ermon, A. Rudra, and C. Ré, "Flashattention: Fast and memory-efficient exact attention with io-awareness," *Advances in Neural Information Processing Systems*, vol. 35, pp. 16 344–16 359, 2022.

Transformers are quadratic in time and memory complexity making them slow and memory hungry on longer sequences. While work has been done in sparse and low-rank approximations to achieve FLOP reduction, Flash Attention makes the algorithms I/O aware. This is important especially when read and write operations are a signification portion of run time. Some such operations include database joins, image processing and numerical linear algebra.

Memory hierarchy and bandwidth can be seen in the left of Figure 2.7. GPU SRAM has the fastest speeds but also least memory followed bu GPU HBM (High Bandwidth Memory) and then CPU DRAM. Reducing the read and writes to the HBM will lead to reduction in wall clock time.

Flash Attention achieves this reduction by applying two techniques:

1. **Tiling** Restructure attention computation by splitting it across blocks, iterate over them and incrementally perform softmax reduction. Softmax reduction is a memory consuming step since it is done by accessing the entire input and performing the reduction. Instead, Flash Attention, through tiling performs it without the whole input.

2. **Recompute** the attention on-chip in the backward pass by storing the softmax normalization factor instead of storing the large intermediate attention matrix.

Through the above optimizations, flash attention requires $\theta(N^2d^2M^{-1}s)$ where N is the sequence length, d the head dimension and M the size of the SRAM with $d \leq M \leq Nd$ instead of the $\theta(Nd+N^2)$ acesses in standard attention.

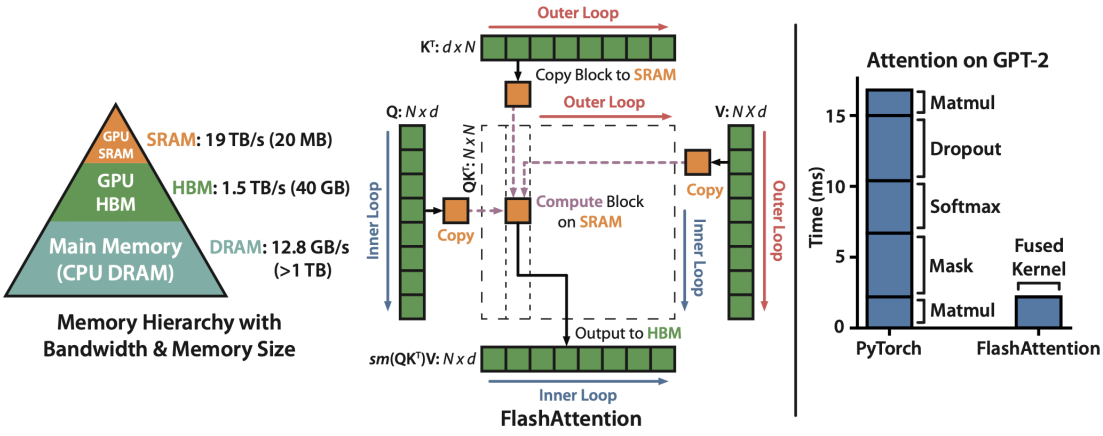


Figure 2.7 Flash Attention

2.2.2 Switch Transformers

[1] T. Dao, D. Fu, S. Ermon, A. Rudra, and C. Ré, "Flashattention: Fast and memory-efficient exact attention with io-awareness," *Advances in Neural Information Processing Systems*, vol. 35, pp. 16 344–16 359, 2022.

Models typically reuse the same parameters for all inputs, Mixture of Expert (MoE) models instead select a different set of parameters for different inputs. This leads to sparsely activated models but with a higher number of parameters. These MoE models also are more complex, have higher communication cost and are in-stable during training. Switch Transformers reduces the communication and computational cost and also mitigates training instability with a simplified routing algorithm and design improvements.

[40] uses a MoE layer for NLP which takes a token representation and routes it to the top-k experts from a set of N experts. Switch Transformer simplified this by routing the token to only a single expert, called the switch layer and showed that the simplification preserved the model quality, reduced routing computation and performed better. For each Switch layer, an auxiliary loss is added to the total model loss during training.

The Switch Transformer is pretrained on the Colossal Clean Crawled Corpus (C4)[36] corpus with a masked language modeling objective. Switch Transformers outperform tuned dense models and MoE Transformers on a speed-quality basis. It also has a smaller computational footprint. The models were also evaluated on downstream tasks. An analysis on scaling and multilinguality was also conducted.

2.3 Documents Understanding

Transformer based LLMs show a quadratic memory growth proportional to the number of tokens in the input. This makes the models unsuitable large text documents. Unlike text documents used in

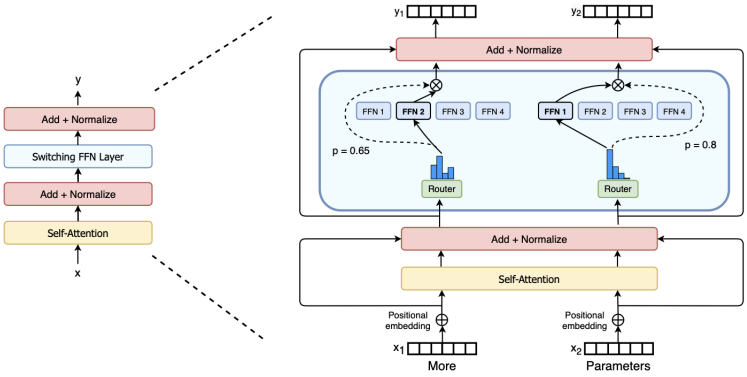


Figure 2.8 Illustration of Switch Transformer Encoder Block

some NLP tasks, physical documents have many more elements. They also contain complex content and layouts, noisy data and fonts. Documents also contain multimodal elements like tables, forms, charts and images. Usually, the text in the documents is connected to these elements.

2.3.1 UniDoc

[1] T. Dao, D. Fu, S. Ermon, A. Rudra, and C. Ré, "Flashattention: Fast and memory-efficient exact attention with io-awareness," *Advances in Neural Information Processing Systems*, vol. 35, pp. 16 344–16 359, 2022.

Documents can be summarized as having the below characteristics:

1. Have a Semantic Structure
2. May contain Semi-Structured Elements like forms and tables
3. May contain images and charts
4. Have a spatial layout

UDoc is a multimodal end-to-end pretraining framework for document images. It extends transformers to train across the various modalities by integrating image information in the pretraining stage through cross-modal interaction between visual and textual information. This is achieved in pretraining through three pretraining objectives viz., Masked Sentence Modeling, Visual Contrastive Learning and Vision-Language Alignment. Figure 2.9 shows the proposed approach.

A document image is fed through an off-the-shelf OCR tool to get the text and bounding boxes. The sentences and Regions of Interest(RoI) are identified. UDoc uses these to extract embeddings using a sentence encoder a visual feature extractor, respectively. These embeddings are then fed into a transformer-based encoder to learn the cross-modal contextualized embeddings that integrate both visual and textual features.

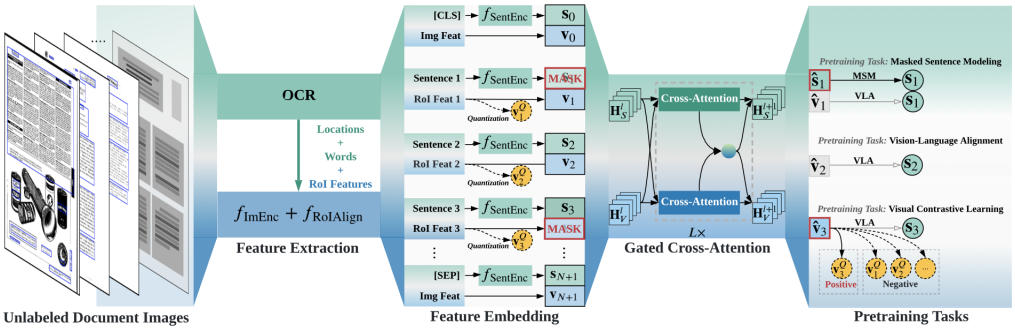
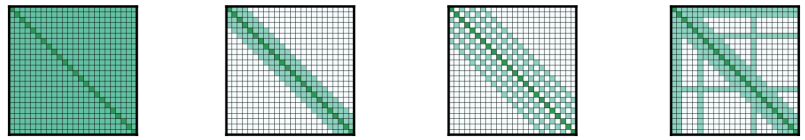


Figure 2.9 UDoc - Approach



(a) Full n^2 attention (b) Sliding window attention (c) Dilated sliding window (d) Global+sliding window

Figure 2.10 Comparison of attention patterns

The pretraining corpus is created based on IIT-CDIP Test Collection 1.0 [41] by removing images from IIT-CDIP present in RVL-CDIP[42]. The sentence encoder is initialized with BERT-NLI-STSb-base. The image encoder is pretrained with a ResNet-50 backbone on the PubLayNet training data[43]. UDoc is then finetuned on Form Understanding, Receipt Understanding, Document Classification and Document Object Detection. The model is evaluated on FUNSD[44] (form understanding task), CORD[45] (receipt understanding task) and RVL-CDIP (document image classification task) datasets. Results show that UDoc outperforms on form and receipt understanding and shows promising results on document image classification.

2.3.2 Other Related Work

Longformer[17] addresses the memory limitation of large context size through a linear scaling attention mechanism by using sliding window, dilated sliding window and global attention patterns as shown in Figure 2.10. LayoutLM[18] has jointly learned on text and image information from documents. LayoutLM adds a 2-D relative position embedding and an image embedding for token images in a document.

2.4 Understanding Graphs

Graphs are non-linearly structured and contain important information in their nodes and edges. Transformers, being inherently linear are limited in their understanding of this non-linearity. Text-

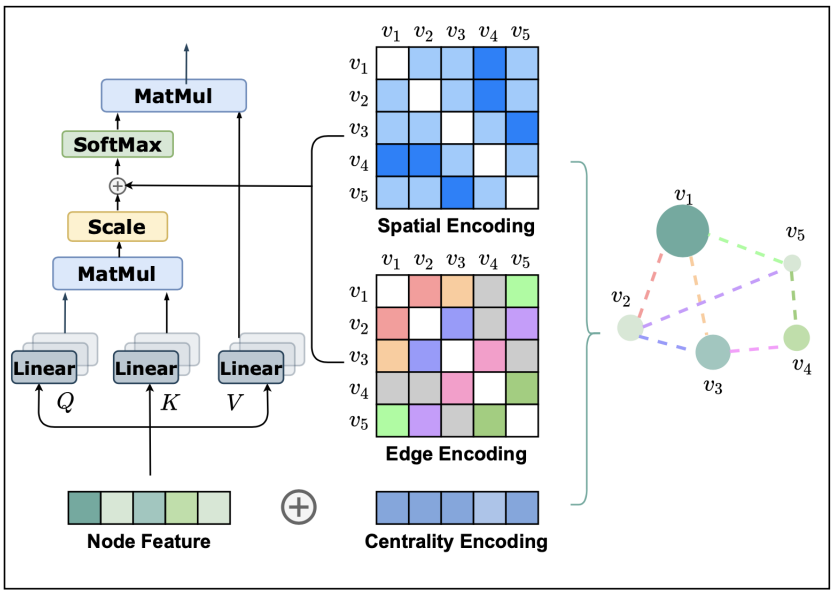


Figure 2.11 Illustration of centrality encoding, spatial encoding and edge encoding in Graphormer

attributed graphs (TAGs) are graphs in which text serves as a node attribute. Legal citation networks and paper citation graphs are good examples of TAGs. Graph Neural Networks (GNNs) used features like skip-gram or bag-of-words for handling text nodes. Current research is focused on using LLMs in conjugation with GNNs to enhance performance on downstream tasks.

2.4.1 Graphormer

[1] Z. Tang, Z. Yang, G. Wang, *et al.*, "Unifying vision, text, and layout for universal document processing," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 19 254–19 264.

Graphormer introduces *centrality encoding* and *spatial encoding* to the transformer model to effectively encode the structural information of a graph. Centrality encoding encodes the importance of a node while spatial encoding adds the structural relations between the nodes. To accommodate the information contained in edges in some graphs, an *edge encoding* is also introduced. Figure 2.11 illustrates the various encoding in Graphormer.

Centrality Encoding A learnable vector is assigned to each node according to its degree (indegree and outdegree) to capture the centrality encoding. The centrality encoding is added to the node features as input.

Spatial Encoding The paper chooses shortest path distance (SPD) between two nodes that are connected as a measure for spatial relation between them. If not they are not connected the SPD is set to -1. A learnable scalar indexed by the nodes is used a bias in attention calculation.

Edge Encoding For each ordered node pair, one of the shortest paths is selected. An averaged of the dot products of the edge features and a learnable embedding is computed along that path. The edge encoding is then added as a bias term to the attention module.

Through the ablations conducted, the results show the effectiveness of spatial encoding, indispensability of centrality encoding and effectiveness of edge encoding to capture spatial information on edges. Graphormer works well on a wide range of benchmarks. The quadratic complexity of the self-attention module which restricts the use of LLMs with large sequences, also restricts the use of Graphormer with large graphs.

2.4.2 GNNs - A Survey

[1] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and S. Y. Philip, "A comprehensive survey on graph neural networks," *IEEE transactions on neural networks and learning systems*, vol. 32, no. 1, pp. 4–24, 2020.

The complex, node-dependent and non-Euclidean nature of graphs has made it challenging to apply existing Machine and Deep Learning algorithms that work well with text and images to graphs. An overview of existing Graph Neural Networks (GNNs) is provided along with a new taxonomy for classifying GNNs, viz. *recurrent GNNs* (RecGNN), *convolutional GNNs* (ConvGNNs), *graph autoencoders* (GAEs), and *spatial–temporal GNNs* (STGNNs).

Recurrent GNNs (RecGNN) aim to learn node representation through RNN architecture with the assumption that a node exchanges information with its neighbours till an equilibrium is reached. The representation is obtained by applying the same set of parameters recurrently over the nodes. Contraction mappings are used to ensure that the recurrent function converges. Most of the early GNNs can be categorized as Rec GNNs.

Convolutional GNNs (ConvGNNs) are very similar to RecGNNs. Instead of using contraction, ConvGNNs use a fixed number of layers with different weights in every layer. ConvGNNs are of two types - Spectral-based and Spatial-based. Spectral-based ConvGNNs approach graph convolution as removing noise from the graph signals whereas Spatial-based approach it as information propagation. Spectral methods assume graphs are undirected.

Graph Autoencoders (GAEs) encode nodes to a latent feature space and then decode the graph from their representation. GAEs can be used to learn low-dimensional network embeddings that contain the nodes' topological information. Since GAEs are able to encode and decode hidden representations, they can be used to generate graphs as well.

Spatial–Temporal GNNs (STGNNs) aim to model dynamic node inputs by capturing the spatial and temporal dependence at the same time. These models learn the temporal representations of the graph structure. Real-time traffic data is an example that can be modeled with STGNNs.

GNNs are used in NLP in text classification, document labeling, sentence generation given a semantic graphs of words and semantic or knowledge graph generation. Though text looks sequential in nature, work has been done to understand internal graph structures such as dependency trees.

2.4.3 Other Related Work

[46] has proposed a CS-TAG dataset which is a diverse collection of eight challenging benchmark datasets for TAGs.

2.5 Legal LLMs and Evaluation

Legal NLP has been gaining more interest in the research community due to the abundance of text documents that are present in the domain. Legal documents in general can be divided into judicial and contractual. These documents differ not only in their content but also in their structure. Results have shown that using out of the box models in the Legal domain has not been satisfactory. This makes exploring models for legal domain lucrative.

2.5.1 Legal BERT

[1] I. Chalkidis, M. Fergadiotis, P. Malakasiotis, N. Aletras, and I. Androutsopoulos, "Legal-bert:" preparing the muppets for court'".," in *EMNLP (Findings)*, 2020, pp. 2898–2904.

While BERT-based models were achieving SOTA performance in many downstream tasks, this paper compared a BERT model with other BERT-based models that were trained on legal domain to evaluate the performance on downstream legal tasks. The results from this paper indicate that the standard finetuning and pretraining guidelines did not generalize well for the legal domain. The following three models are compared:

1. BERT model (out of the box)
2. Continued pretraining on BERT with legal data
3. Pretrained BERT from scratch on legal data

The models were trained with data from EU Legislation, UK Legislation, European Court of Justice (ECJ) cases, European Court of Human Rights (ECHR) cases, US court cases and US contracts. The models were evaluated on text classification and sequence tagging using EURLEX57K[47], ECHR-CASES[48] and CONTRACTS-NER[49] datasets. The results showed that the performance gains were higher in the domain specific models in the more challenging tasks (viz., multi-label classification in ECHR-CASES and contract header, lease details in CONTRACTS-NER) where in-domain knowledge is more important.

2.5.2 Lawformer

[1] C. Xiao, X. Hu, Z. Liu, C. Tu, and M. Sun, "Lawformer: A pre-trained language model for chinese legal long documents," *AI Open*, vol. 2, pp. 79–84, 2021.

Dataset	Source	Sub-domain	Task Type	Training/Dev/Test Instances	Classes
ECtHR (Task A)	Chalkidis et al. (2019a)	ECHR	Multi-label classification	9,000/1,000/1,000	10+1
ECtHR (Task B)	Chalkidis et al. (2021c)	ECHR	Multi-label classification	9,000/1,000/1,000	10+1
SCOTUS	Spaeth et al. (2020)	US Law	Multi-class classification	5,000/1,400/1,400	14
EUR-LEX	Chalkidis et al. (2021a)	EU Law	Multi-label classification	55,000/5,000/5,000	100
LEDGAR	Tuggener et al. (2020)	Contracts	Multi-class classification	60,000/10,000/10,000	100
UNFAIR-ToS	Lippi et al. (2019)	Contracts	Multi-label classification	5,532/2,275/1,607	8+1
CaseHOLD	Zheng et al. (2021)	US Law	Multiple choice QA	45,000/3,900/3,900	n/a

Figure 2.12 Summary of LexGLUE datasets

Lawformer is a Longformer-based model for Chinese Long Legal Document Understanding. The model was created to accommodate the longer token length requirements of legal documents. The model uses local sliding window attention and the global task motivated full attention to capture the long-distance dependency. Thel mode was trained with the Masked Language Modeling (MLM) objective by continuing pretraining from the *RoBERTa-wwm-ext* checkpoint.

The model was evaluated on judgement prediction, similar case retrieval, legal reading comprehension and legal question answering. While the results are comparable with RoBERTa and L-RoBERTa, the gains are not always very significant. Tasks that require more complex reasoning with an understanding on the domain did not achieve promising results.

2.5.3 LexGLUE

[1] I. Chalkidis, A. Jana, D. Hartung, *et al.*, "LexGLUE: A benchmark dataset for legal language understanding in English," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 4310–4330. DOI: 10.18653/v1/2022.acl-long.297.

Following in the lines of the SuperGLUE[37], LexGLUE was created as a Legal General Language Understanding Evaluation benchmark. It is a collection of seven existing English Legal NLP datasets from the United States and the European Union. It comprises of ECtHR task A[48] and B[50], SCOTUS[51], EURLEX[52], LEDGAR[53] and UNFAIR-ToS[54], which are all Multi-label Classification Datasets. The benchmark also has CaseHOLD [55] which is a multiple choice QA dataset. A summary of the datasets can be seen in Figure 2.12

The European Court of Human Rights (ECtHR) [48] hears allegations that a state has breached human rights provisions of the European Convention of Human Rights (ECHR). The input to this is a list of facts of a case and output is a set of violated articles in Task A. In Task B, the input remains the same but the output is a set of allegedly violated set of articles. The label set includes 10 ECHR articles.

The US Supreme Court (SCOTUS) [51] is the highest court in the US and usually hears only the cases which have not been sufficiently resolved by lower courts. The dataset contains SCOTUS opinions combined with the Supreme Court Database(SCDB). SCDB is used to classify the opinions into 14 issue areas. This is a single-label multi-class classification task.

EUR-Lex [52] portal contains the European Union (EU) legislation. This task takes a document as input and predicts its corresponding EuroVoc[1] labels. The evaluation benchmark contains the top 100 most frequent concepts/classes for classification. The task is to predict the EuroVac concepts for a given input document.

LEDGAR (Labeled EDGAR) [53] is a dataset for contract provision/paragraph classification. The contracts in this dataset come from the US Securities and Exchange Commission (SEC) fillings[2]. For the benchmark, 100 most frequent categories from the 12.5k provisions are used. This is also a single-label multi-class classification task.

The UNFAIR-ToS dataset [54] contains 50 Terms of Service(ToS) from online platforms. The dataset is annotated at sentence level with 8 types of unfair contractual terms that violate user rights according to the EU consumer law. The task takes a sentence as input and gives a set of unfair laws, if any, as output.

The CaseHOLD (Case Holdings on Legal Decisions) dataset[55] contains multiple choice questions about US Court Case Holdings from the Harvard Law Library case law corpus. Holdings are summaries of legal ruling that accompany referenced decisions relevant relevant to a case. The input is an excerpt from a court decision, the output is the correct holding related to the particular reference, given five choices of holdings.

2.5.4 Other Related Work

The LeXFiles[56] and Pile of Law [57] are more recent corpora that contain 19B and 32B tokens respectively. While Pile of Law corpus contains majority of the data from the United States, LeXFiles contains data from 6 primarily English-speaking legal systems - European Union, Council of Europe, Canada, United States of America, Canada and India. Alongside LeXFiles, a knowledge probing benchmark LegalLAMA was also released.

There has been recent work in the Indian Legal domain as well. ILDC(Indian Legal Documents Corpus)[24] is a dataset of 35k Indian Supreme Court Cases annotated with the original court decision. HLDC(Hindi Legal Documents Corpus)[25] is a corpus of 900k Hindi Legal Documents, collected from the district courts of Uttar Pradesh. MILDSum[26] is a benchmark dataset for Multilingual Summarization of Indian Case Judgements. The dataset contains 3122 judgements with their summaries in English and Hindi. InLegalBERT and InCaseLawBERT[23] are an attempt at a BERT-based model for Indian Legal Data created by continuing pretraining on LegalBert and CaseLawBERT using the data from 5.4M Indian Legal Documents.

[1] Multilingual Thesaurus maintained by the Publications Office

[2] publicly available from EDGAR10 (Electronic Data Gathering, Analysis, and Retrieval system)

Chapter 3

Current Progress

The courts in India follow a hierarchy and are categorized into Supreme, High and District and Subordinate courts as shown in 3.1. Usually, cases start at the lower courts and then can be appealed at higher courts. Supreme Court and High Court proceedings are in English whereas the other lower court proceedings may be in local languages. Cases in the Indian Judicial system can cite Precedents, Acts, Customs and in some rare cases Precedents from High Courts. The current research is limited to only citations involving case precedents.

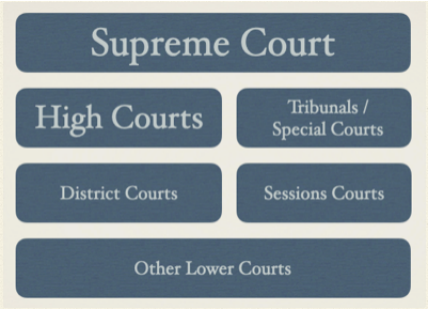


Figure 3.1 Court Hierarchy in Indian Judiciary

In the current work, the task of Legal Precedent Generation is formulated as generating paragraphs that can be used for citations, given the current context of a case. The aim is to achieve this by retrieving relevant cases and then generating a paragraph containing the information from the precedents. Identifying precedents is a non-trivial task considering the language, number of past judgements, corresponding case documents and their length. A precedent citation, usually refers to a specific paragraph from the judgement it is citing and not the whole case. Case documents are not always structured enough to identify paragraph boundaries.

The work requires downloading the documents from Supreme Court, 25 High Courts and about 18k[1] District and Subordinate Court Complexes. According to NJDG, there are 44M pending cases and 159M

[1] https://doj.gov.in/the-national-judicial-data-grid-njdg/

disposed cases at district courts[2], 6M pending and 37M disposed cases at High Courts[3] and 79k pending and 45k disposed cases at Supreme Court[4].

Data Collection

Currently, all the judgements from the Supreme Court of India and about 30k judgements from various High Courts have been downloaded as pdfs and converted to text. 1.5M other court documents such as Orders have also been downloaded from the Supreme Court. Documents from about 500 district and subordinate complexes across the states of Andhra Pradesh, Telangana and Tamil Nadu have also been downloaded as pdfs. The subordinate courts in the states of Andhra Pradesh and Telangana function solely in English. The court documents from Tamil Nadu are in both English and Tamil.

The supreme court documents were downloaded from the Supreme Court of India website[5]. These documents were converted into plain text. Information regarding the Petitioner, Respondent, Judgement Date and Bench has also been collected.

Downloading the district court data is ongoing. The documents are being downloaded from ecourts[6]. District documents contain some scanned documents and in some cases handwritten scanned documents. Some of the documents also contain handwritten forms. District courts in India are allowed to function in a local language, leading to multilingual documents. Converting these documents into text is also ongoing.

[2] https://njdg.ecourts.gov.in/njdgnew/
[3] https://njdg.ecourts.gov.in/hcnjdgnew/
[4] https://njdg.ecourts.gov.in/scnjdg/
[5] https://www.main.sci.gov.in/judgments
[6] https://districts.ecourts.gov.in

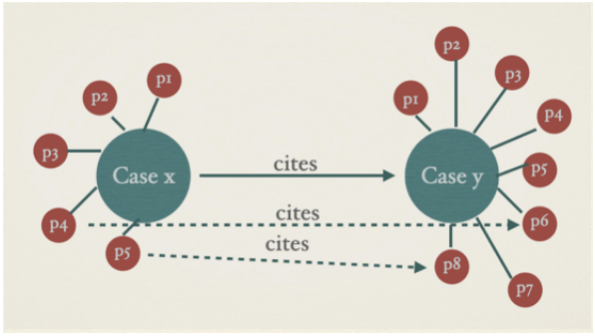


Figure 3.2 Depiction of Citation Network

Citation Network

The citations from cases were extracted to build a citation network. The documents were split into paragraphs to bring a paragraph-level granularity to the network. The underlying structure is a heterogeneous graph with the documents and paragraphs as nodes. The network is disjointed as not all nodes are connected. A snapshot of the graph with only document nodes is shown in Figure3.3. The citations for each case were extracted to create a citation list.

27k cases cite precedents while 32k cases are being cited. A few cases are cited more often than others and cases cited a minimum of 1 case with some landmark cases citing a maximum of 205 cases. On an average, each Supreme Court judgement cited 4.7 precedents.

A part of the current work was presented at the 19th International Conference on Artificial Intelligence and Law (ICAIL) 2023, Doctoral Consortium in June[7].

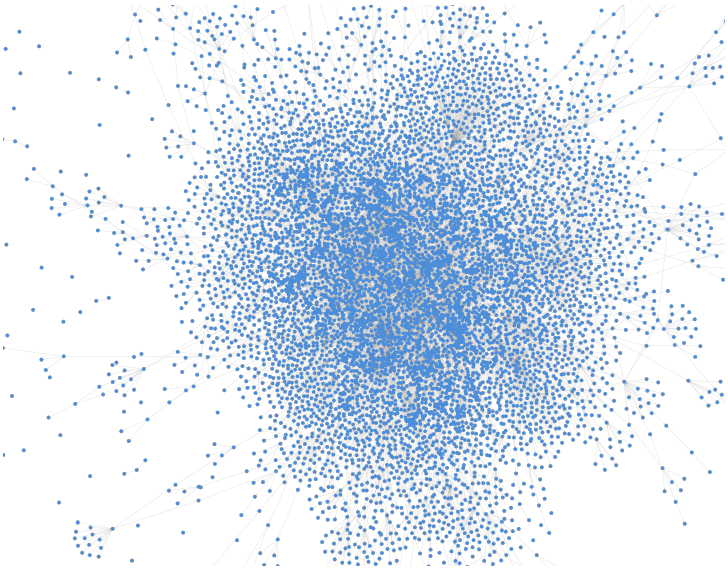


Figure 3.3 Legal Citation Network with Case Documents as Nodes

[7] https://icail2023.di.uminho.pt

Chapter 4

Conclusion and Future Work

The report has gone over the some of the current research in LLMs, Graphs, Document Understanding and Legal NLP. Some limitations and some solutions to those limitations have also been explored. NLP for the legal domain, LLMs with Graphs and efficiently encoding large documents remain active areas of research.

Next Steps

1. Data Acquisition: Download the judgements from all the 25 high courts and 758 districts (Each district has several court establishments)
2. Benchmark Dataset(s): There is a need to create an evaluation benchmark dataset containing several tasks in the Indian Legal domain to better understand model performance.
3. Legal LLMs on Indian Judicial Data: Carry pretraining(continuing/from scratch) for use in downstream tasks.
4. Citation prediction using the citation network and text from the nodes.
5. Multilingual LLMs and Citation Prediction: Expand work to accommodate the multilingual documents from district courts.

Chapter 5

List of Selected Papers

General LLMs Related

[1] J. Wei, X. Wang, D. Schuurmans, *et al.*, "Chain-of-thought prompting elicits reasoning in large language models," *Advances in Neural Information Processing Systems*, vol. 35, pp. 24 824–24 837, 2022.

[2] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," *Advances in neural information processing systems*, vol. 35, pp. 22 199–22 213, 2022.

[3] W. Fedus, B. Zoph, and N. Shazeer, "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity," *The Journal of Machine Learning Research*, vol. 23, no. 1, pp. 5232–5270, 2022.

[4] T. Dao, D. Fu, S. Ermon, A. Rudra, and C. Ré, "Flashattention: Fast and memory-efficient exact attention with io-awareness," *Advances in Neural Information Processing Systems*, vol. 35, pp. 16 344–16 359, 2022.

[5] B. Lester, R. Al-Rfou, and N. Constant, "The power of scale for parameter-efficient prompt tuning," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021, pp. 3045–3059.

Document Understanding

[1] J. Gu, J. Kuen, V. I. Morariu, *et al.*, "Unidoc: Unified pretraining framework for document understanding," in *Advances in Neural Information Processing Systems*, M. Ranzato, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, Eds., vol. 34, Curran Associates, Inc., 2021, pp. 39–50.

Graphs

[1] C. Ying, T. Cai, S. Luo, *et al.*, "Do transformers really perform badly for graph representation?" *Advances in Neural Information Processing Systems*, vol. 34, pp. 28 877–28 888, 2021.

[2] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and S. Y. Philip, "A comprehensive survey on graph neural networks," *IEEE transactions on neural networks and learning systems*, vol. 32, no. 1, pp. 4–24, 2020.

Legal

[1] I. Chalkidis, M. Fergadiotis, P. Malakasiotis, N. Aletras, and I. Androutsopoulos, "Legal-bert:" preparing the muppets for court'".," in *EMNLP (Findings)*, 2020, pp. 2898–2904.

[2] C. Xiao, X. Hu, Z. Liu, C. Tu, and M. Sun, "Lawformer: A pre-trained language model for chinese legal long documents," *AI Open*, vol. 2, pp. 79–84, 2021.

[3] I. Chalkidis, A. Jana, D. Hartung, *et al.*, "LexGLUE: A benchmark dataset for legal language understanding in English," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 4310–4330. DOI: 10.18653/v1/2022.acl-long.297.

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